

DAIVIK WADHWANI

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Summary

Full Stack engineer (React, Node.js, MongoDB) with production experience across MERN systems and serverless AWS architecture. Co-authored an HCI research paper on VR accessibility, accepted to ACM ASSETS 2026.

Education

Indian Institute of Information Technology Sricity

August 2023 - June 2027

Bachelor of Technology in Computer Science (CGPA of 7.76)

Experience

Research Intern, HCI Lab

Jan 2026 – Present

Indian Institute of Information Technology Sricity

- Co-authored a research paper on one-handed VR accessibility for players with motor disabilities, accepted to ACM ASSETS 2026, the ACM SIGACCESS Conference on Computers and Accessibility.
- Designed and ran a 17-participant user study applying NASA-TLX (Raw TLX 42.0) and System Usability Scale (65.8) instruments; identified a negative learning slope in 12 of 13 long-session participants as quantitative evidence of motor adaptation.
- Engineered the evaluated system in Unity using XR Interaction Toolkit and OpenXR, implementing position mirroring, freeze-offset, and saber-swap modes to enable full gameplay from a single controller.

Projects

SplitSmart | MERN, Socket.io, Gemini Vision API, JWT

2026

AI-powered group expense splitter with real-time cooperative bill claiming and OCR receipt parsing.

- Engineered a 3-phase greedy minimum cash flow algorithm reducing N debts to at most N-1 transactions; integrated Gemini 2.5-Flash via zero-disk in-memory pipeline for receipt OCR; implemented OCC with jitter-based retry backoff to resolve simultaneous item-claim race conditions across Socket.io clients.
- Optimised query performance with compound MongoDB indexes across 4 models, cutting expense doc scans from 6,021 to 15 and raw DB time from 9ms to 1ms; hardened API with role-based JWT refresh flow, sparse unique indexing for frictionless guest onboarding, and rate limiting to prevent token exhaustion.

Bookish | MERN, Redux Toolkit, Stripe, Upstash Redis, Jest

2026

Full-stack book marketplace connecting buyers, sellers, and admins with role-based dashboards.

- Integrated Upstash Redis cache-aside layer across 11 critical endpoints achieving 84.82% average latency reduction — warm cache hits consistently under 30ms, down from 120-230ms cold DB access.
- Implemented CI/CD pipeline with GitHub Actions running 48 Jest tests across 12 suites on every push; added B-Tree indexing, field projections, and `.lean()` hydration bypass reducing CPU overhead by 35-50%.

PowerPulse | AWS Lambda, DynamoDB, API Gateway, Cognito, Serverless

2025

Serverless energy analytics system with rule-based and AI-driven insights.

- Architected a fully serverless backend on AWS Lambda, API Gateway, Cognito, and DynamoDB — eliminating idle infrastructure cost and enabling automatic scaling without capacity planning.
- Devised caching strategy that cut AI response time from 1800ms to 200ms (89% faster) and reduced API costs by 70%; integrated Hugging Face inference with fallback control.

Technical Skills

Languages: C, C++, C#, Java, JavaScript, HTML, CSS

Frontend: React, Tailwind CSS, Redux Toolkit, TanStack Query

Backend: Node.js, Express.js, REST APIs, Socket.io, JWT, Multer

Cloud & Infra: AWS Lambda, API Gateway, Cognito, Cloudinary, Render, Vercel

Databases: MongoDB, MySQL, DynamoDB, Upstash Redis

Tools: Git, GitHub, Jest, GitHub Actions, Unity, OpenXR, XR Interaction Toolkit, Postman

Achievements

- Secured 3rd place at Global Game Jam 2025 — shipped a complete game under a 48-hour constraint.
- Rated 1580 on LeetCode; solved 250+ problems across LeetCode and Codeforces.
- XR Lead, IOTA Club — leading XR initiatives, mentoring members, and managing lab projects.